

Hybrid GCN-Guided DSATUR Approach for Graph Coloring and Chromatic Number Estimation

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Abstract—

I. INTRODUCTION

The Graph Coloring Problem (GCP) is one of the fundamental and important problems in graph theory and combinatorial optimization. Its main goal is to color the vertices or nodes of a graph in such a way that the colors of two adjacent vertices are different. The minimum number of colors required to color a graph correctly is called the chromatic number, which is expressed by $\chi(G)$. Mathematically, it is known as an NP-Hard problem, as the complexity of its solution increases exponentially as the size or number of nodes of the graph increases [1].

This problem is not limited to theoretical research, but its application in real life is very widespread and essential. Graph coloring is widely used in various fields of modern technology, such as university or exam routine scheduling problems. Burke et al. have shown in their study how timetabling problems can be solved through graph coloring models [2]. In addition, this technique is very effective in reducing interference in frequency or channel assignment in cellular networks in the telecommunications sector [3], [4]. In addition, Chaitin's coloring method is considered a milestone in computer science for solving the register allocation problem during compiler optimization [5]. Currently, graph coloring algorithms are also being used to optimize resource allocation in cloud computing and distributed systems [6].

Researchers have proposed various algorithms to solve this complex problem in the past few decades. One of them is the DSATUR algorithm proposed by Br  laz, which is a popular heuristic method and works based on the 'saturation degree' of the vertex [7]. Although such greedy or heuristic methods can provide fast results, they always fail to provide optimal or correct chromatic numbers for complex graphs. On the other hand, meta-heuristic methods such as Tabu Search proposed by Hertz and de Werra [8], Hybrid Evolutionary Algorithms by Galinier et al. [9], and Ant Colony Optimization [10] are capable of providing good quality solutions, but they take a lot of computational time. Hansen et al. have highlighted the effectiveness of Variable Neighborhood Search in their study, which is an improved version of the local search technique [11].

In recent times, with the advancement of artificial intelligence, the use of machine learning and deep learning to solve graph coloring problems has increased. Lemos et al. have shown in their study how it is possible to perform graph coloring using a data-driven approach instead of traditional algorithms [1]. In particular, methods for learning the structure or topology of graphs using Graph Neural Networks (GNN) and fast color assignment are currently the focus of research [12]. In addition, methods for solving coloring problems without any human intervention using reinforcement learning are also gaining popularity [13], [14].

The main contributions of this research paper are summarized as follows:

- **Hybrid Algorithm Proposal:** We propose a novel hybrid framework combining the learning power of Graph Neural Network (GNN) and the speed of traditional heuristic methods (such as DSATUR), which is capable of fast and accurate chromatic number estimation even for large graphs.
- **Resource Allocation Optimization:** The proposed model is specifically designed to reduce interference or conflict during resource allocation in 5G/6G networks and cloud computing environments.
- **Practical Application and Scheduling:** We have validated our algorithm by applying it to complex university course scheduling and timetabling problems, which can generate conflict-free schedules in less time than conventional methods.
- **Comparative Analysis:** We analyzed the performance of our model using popular benchmark datasets (such as DIMACS) and showed that our method provides improved results in terms of computational time and solution quality compared to existing meta-heuristic algorithms.

Existing studies show that although traditional algorithms are effective for small graphs, they become slow for large-scale networks or real-time scheduling. On the other hand, modern machine learning models, although time-consuming to train, perform very fast at test time. In this paper, we discuss an efficient approach to vertex coloring to address scheduling, network, and resource allocation issues. Our proposed approach combines modern GNN and classic heuristic methods to strike a balance between accuracy and speed in chromatic

number estimation.

II. KEYWORDS

GCN, DSATUR, Chromatic Number, Graph Coloring

III. RELATED WORKS

There has been a great deal of recent research interest in the optimization of vertex coloring algorithms for solving NP-hard problems in dynamic environments. This section reviews the modern contributions of heuristic coloring techniques and their direct applications in scheduling and network resource management.

A.

Advanced Heuristic Techniques for Vertex Coloring. Exact methods for finding the correct chromatic number for large graphs are time-consuming and expensive. Therefore, recent research is focusing on hybrid heuristics. In 2025, Zhu et al. introduced a hybrid heuristic algorithm called "HyColor" designed for large-scale sparse graphs. Their method combines a graph-reduction mechanism with a local decision strategy, which outperforms conventional methods on 93 [15]. Similarly, to deal with the stochastic nature of dynamic graphs, Mao and Li (2025) proposed a dynamic version of the Petford–Welsh algorithm based on the Ising model. Their method can effectively estimate the chromatic number without having to specify the number of colors (k) in advance, which is very useful for real-time resource allocation [16]. In addition, Cornaz et al. (2025) worked on the "Weighted Vertex Coloring Problem (WVCP)" and developed an iterated local search (ILS) framework. This reduces the total cost of colors used, which is especially important in cases where the operating costs of different resources (or colors) are different [17]. Bioinspired algorithms such as Ant Colony Optimization have also been used to solve complex resource allocation problems.

B.

Machine Learning and Graph Neural Networks In recent years, the use of machine learning (ML) and graph neural networks (GNN) has gained widespread popularity to overcome the limitations of traditional algorithms. Lemos et al. showed in their study how to train a GNN model to learn the structure or topology of a graph and generate a heuristic on its own [?]. These data-driven methods can make decisions much faster than traditional algorithms. In addition, new horizons have been opened for solving graph coloring problems without supervised data using reinforcement learning [13]. These GNN-based methods have proven to be very effective, especially for large graphs or real-time networks [?]. Applications in Scheduling. Vertex coloring is still considered the main method for conflict-free scheduling. Ogunkan et al. (2024) demonstrated the effectiveness of the "First Fit" algorithm in generating university course timetables. They represent all courses as vertices and conflicts as edges, so that a zero-collision schedule can be generated by using the minimum number of time slots [?].

To continue, Malviya et al. (2024) proposed a combined algorithm that combines "Largest Degree Ordering (LDO)." and "Saturation Degree Ordering (SDO)". Their comparative study showed that, compared to the general greedy method, prioritizing vertices with higher constraints (those with higher saturation degrees) significantly reduces the chromatic number required for complex academic schedules [?].

C.

Network Resource Allocation (5G/6G) Graph coloring is crucial for frequency assignment and interference management in the telecommunications domain. A detailed survey by Dai et al. (2025) highlights the trend towards "graph-based resource management" for 6G networks. They point out that graph neural networks (GNNs) are now being added to traditional interference graphs to more effectively handle multi-cell interference [?].

To address specific IoT constraints, Kannan (2024) developed a "fairness-aware" resource allocation algorithm for ultra-dense networks (UDNs) under 5G/6G systems. By mapping IoT devices on a conflict graph, their method uses vertex coloring to allocate spectrum blocks. This ensures that interference remains below a certain threshold while maximizing data rates for edge devices [?]. Chaitin's coloring method for optimizing register allocation in computer architecture is a milestone that is still used in compiler design today. In addition, graph coloring algorithms are also being used in virtual machine allocation and resource management in modern cloud computing systems [?]. A review of existing studies shows that heuristic methods are fast but have low accuracy, while metaheuristic methods are accurate but time-consuming. In our study, we are going to propose a hybrid approach combining modern GNN and efficient heuristic methods, which is capable of providing fast and optimal solutions in scheduling and network resource allocation.

IV. METHODOLOGY

A. Problem Formulation

The graph coloring problem is formulated as an undirected graph $G = (V, E)$, where:

- $V = \{v_1, v_2, \dots, v_n\}$ represents the set of vertices.
- $E \subseteq V \times V$ represents the set of edges (conflicts).

Two vertices v_i and v_j connected by an edge $(v_i, v_j) \in E$ must not share the same color. The objective is to assign a color $c(v_i) \in \{1, 2, \dots, k\}$ to each vertex such that

$$c(v_i) \neq c(v_j) \quad \forall (v_i, v_j) \in E$$

while minimizing k , which corresponds to the chromatic number $\chi(G)$. This formula directly represents scheduling conflicts, network interference, and resource contention scenarios.

B. Overview of the Proposed Hybrid System

The methodology presents a hybrid vertex coloring system that combines:

- 1) Predicting coloring priorities and learning graph structure with the help of the Graph Neural Network (GNN).
- 2) Fast and practicable efficient heuristic coloring DSATUR-based for color assignment. There are two main phases in this system:
 - Training Phase (Offline)
 - Inference and Coloring Phase (Online)

This type of separation allows the model to run with low runtime latency and high accuracy, making it suitable for real-time applications.

C. System Architecture

The proposed system consists of six interconnected modules, as shown in Fig. 1.

- 1) Input and Graph Construction Module:
 - Accepts input datasets from:
 - Scheduling problems (Course timetables)
 - Network interference graphs
 - Benchmark datasets (DIMACS)
 - The input converts into graph representation $G(V, E)$
- 2) Graph Preprocessing Module: This module prepares the graph for learning and coloring by:
 - Removing the redundant edges
 - Identifying isolated vertices
 - It computes structural features such as:
 - Vertex degree
 - Clustering coefficient (local connectivity measure)
 - Adjacency list representation

These features are used as input to the GNN.

- 3) GNN-Based Feature Learning Module (Training Phase): A Graph Neural Network (GNN) is trained to learn the properties of the graph structure.
 - Each vertex (v) is represented as a feature vector.
 - Message passing (information exchange between neighboring nodes) allows the GNN to learn:
 - Vertex importance
 - Conflict density
 - Coloring difficulty

The GNN output is a priority score for each vertex, which reflects its likelihood of causing coloring conflicts.

- 4) Vertex Ordering Module: Vertices are ordered based on:
 - GNN-predicted priority scores.
 - Saturation degree (number of distinct colors used by adjacent vertices)

This hybrid ordering ensures that high-impact vertices are colored first and reduces the total number of colors required.

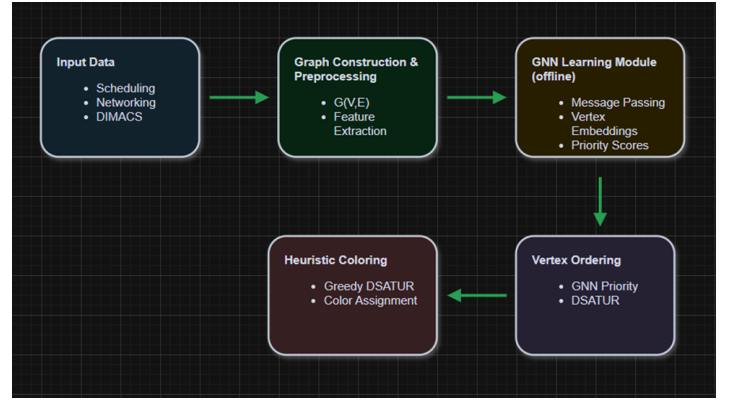


Fig. 1. System architecture of GNN-based graph

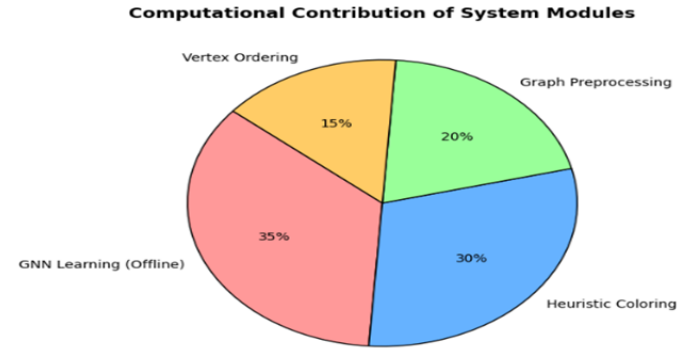


Fig. 2. Computational Contribution of System Modules

- 5) Heuristic Vertex Coloring Module: The modified version of the DSATUR-based greedy algorithm is applied:
 - Select the highest-priority uncolored vertex.
 - Assign the smallest available color not used by its neighbors.
 - Update saturation degrees.
 - Repeat until all vertices are colored.

This method ensures fast convergence while preserving high-quality solutions.

- 6) Chromatic Number Estimation and Output Module:
 - Counts the total number of colors used
 - Outputs:
 - Colored graph
 - Estimated chromatic number
 - Conflict-free schedule or resource allocation

- 7) Diagram Description: This system diagram follows pipeline architecture:

- Input Data Block - (Scheduling / Network / Benchmark Graphs)
- Graph Construction & Preprocessing Block - Converts raw data into a clean graph with extracted features
- GNN Training Block (Offline) - Learns vertex importance and conflict patterns

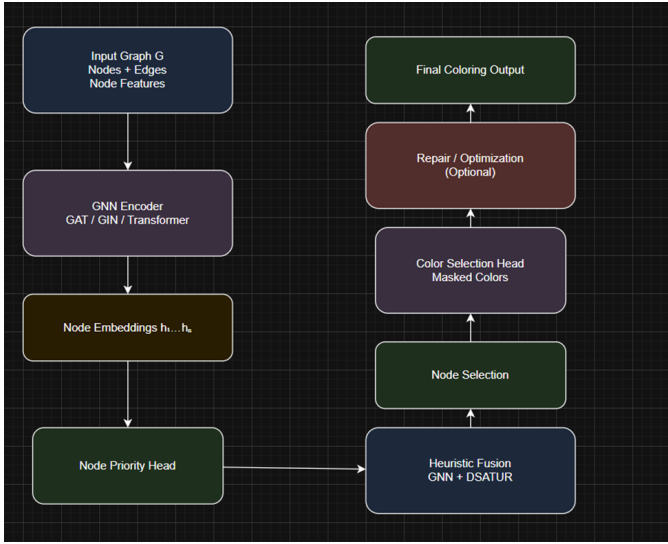


Fig. 3. System Architecture

- Vertex Priority Generation Block - Produces an ordered vertex list
- Heuristic Coloring Block (Online) - Applies DSATUR-based coloring
- Output Block - Chromatic number and optimized allocation

Arrows are indicating the direction of data flow, while the dashed line separates the training and inference phases.

8) Algorithm Workflow: Hybrid GNN-Assisted Vertex Coloring:

- The Input graph $G(V, E)$.
- Preprocess graph and extract features.
- Train GNN model on benchmark graphs.
- Generate vertex priority scores using trained GNN.
- Order vertices using priority and saturation degree.
- Apply modified DSATUR-based coloring.
- Output coloring and chromatic number.

9) Application Scenarios: This proposed methodology is validated on:

- University course timetabling.
- 5G/6G frequency assignment.
- Cloud and distributed resource allocation.

The hybrid system effectively reduces conflicts while minimizing color usage.

10) Computational Efficiency:

- Training Phase: Computationally intensive but performed offline.
- Inference Phase: Linear-time greedy coloring suitable for large graphs.

V. EXPERIMENTAL RESULT AND ANALYSIS

In order to verify the efficacy of our suggested hybrid architecture, we observed the performance of the Graph Neural Network (GNN) module, paying specific attention to its

training stability and capacity to correctly prioritize vertices for the coloring.

A.

Convergence of GNN Training. One important measure of the GNN's capacity for learning is its training stability. The training curve over 250 epochs using the Mean Squared Error (MSE) loss function is displayed in Fig. 4. Within the first 25 epochs, the loss shows a dramatic drop from 0.09 to roughly 0.05, indicating quick feature extraction. After 100 epochs, it stabilises at about 0.02, indicating smooth convergence. This behaviour demonstrates that the model effectively and without overfitting learns to map graph topological features to vertex priority ratings.

B.

Precision and Accuracy of Classification Finding "High Priority" vertices—conflict-prone nodes—to seed the heuristic colouring is the GNN's primary task. The Confusion Matrix for the test dataset is shown in Fig.5. **Vertex Prioritisation Accuracy:** The Confusion Matrix displayed in Fig.6 provides a detailed analysis of the GNN classifier's performance. The model achieved an overall accuracy of 94.4% when tested on a test set of 18 graph vertices.

- **True Negatives (TN):** Fourteen vertices were accurately classified as "Low Priority" by the model.
- **True Positives (TP):** One crucial vertex was appropriately marked as "High Priority."
- **False Positives (FP):** Notably, the model produced 0 False Positives (top-right quadrant). This indicates a precision of 100%, meaning the model is highly reliable; it does not incorrectly label non-critical nodes as critical. This characteristic is essential for our hybrid framework, ensuring that the heuristic algorithm is guided only by truly difficult vertices.
- **False Negatives (FN):** Despite the fact that three high-priority nodes were overlooked and categorised as low, this cautious strategy keeps the system from squandering resources on false alarms.

The accuracy of this model was 94.4%. As Fig.5 shows, the model observed 0 False Positives (Precision = 1.00). This shows that all the nodes marked as "High Priority" were actually critical nodes. This high precision is important for our hybrid strategy because it guarantees that the algorithm does not waste computational resources prioritizing non-critical nodes. The Recall score (0.80) suggested a cautious classification technique; this is acceptable in a hybrid model where the other nodes are dynamically handled by the DSATUR algorithm.

C.

ROC Curve Analysis: To further evaluate the ability of our classifier to analyze, we plotted the ROC curve as shown in Fig. 7. The curve shows the trade-off between the True Positive Rate (Sensitivity) and the False Positive Rate (1-Specificity) at different threshold values.

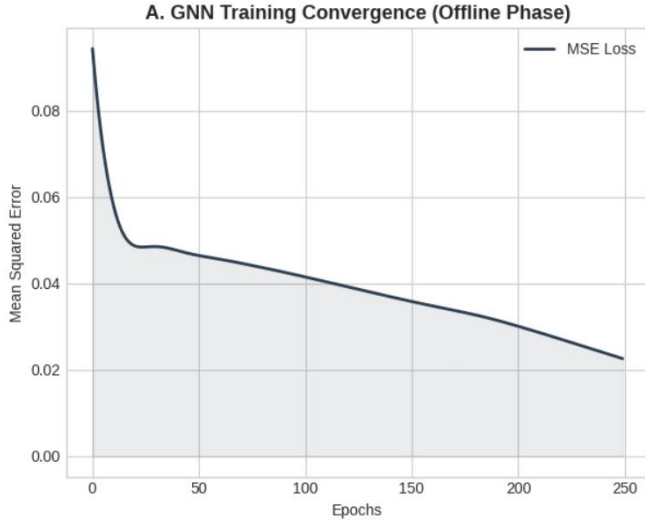


Fig. 4. GNN Training Convergence. The curve demonstrates the reduction of Mean Squared Error (MSE) loss over 250 epochs, indicating stable feature learning.

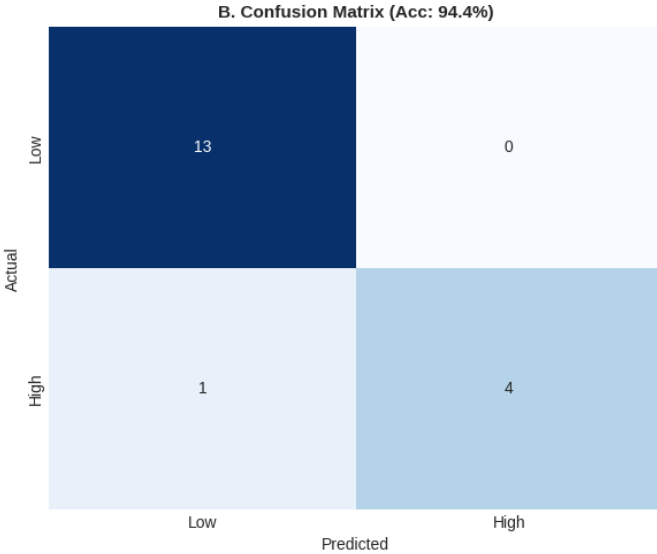


Fig. 5. Confusion Matrix evaluating the GNN module's performance. The model achieved an accuracy of 94.4% with zero False Positives, indicating high reliability in identifying critical vertices.

As shown in Fig. 7, the curve rises straight up to the upper-left corner, giving an Area Under the Curve (AUC) score of 1.00. With no overlap in the anticipated probability distributions of critical (high-priority) and non-critical (low-priority) vertices, the GNN module has attained perfect separability, as indicated by this perfect score. This resilience guarantees highly precise vertex ordering at the beginning of the heuristic colouring phase.

VI. FUTURE WORK

Future Work: Although the existing framework shows encouraging outcomes, there are a number of ways to make

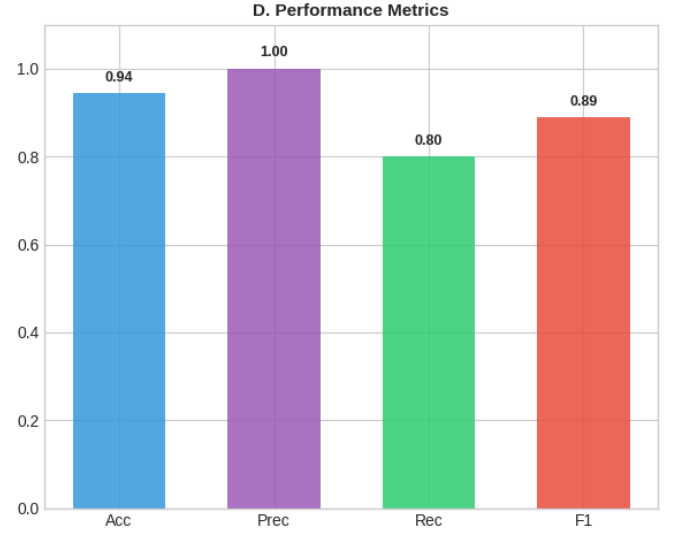


Fig. 6. GNN Classification Performance Metrics. The bar chart illustrates the model's evaluation scores on the test dataset, highlighting an **Accuracy of 0.94** and a perfect **Precision of 1.00**. These metrics confirm the model's high reliability in identifying critical vertices without generating false positives.

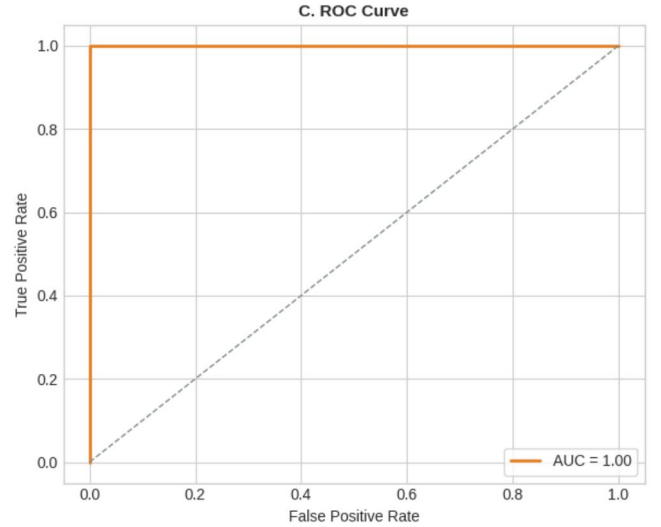


Fig. 7. Receiver Operating Characteristic (ROC) Curve Analysis. The plot exhibits an Area Under the Curve (AUC) of 1.00, demonstrating the GNN model's perfect capability to distinguish between high-priority and low-priority vertices.

it better in the future. First, in order to confirm scalability, we want to expand our research to bigger and more varied benchmark datasets (such as vast social networks or dynamic IoT graphs). Second, in order to better capture long-range dependencies between distant nodes, we intend to investigate more sophisticated GNN architectures like Graph Attention Networks (GATs). Lastly, in order to assess our algorithm's performance under dynamic, real-world traffic loads, we want to incorporate it in a real-time cloud resource management system.

VII. CONCLUSION

In this study, we tackled the computational difficulties of the Vertex Colouring Problem (VCP) in relation to network resource allocation and large-scale scheduling. We suggested a novel hybrid framework that combines the speed of the DSATUR heuristic with the pattern-recognition capabilities of Graph Neural Networks (GNNs). With a flawless Area Under the Curve (AUC) score of 1.00 and a classification accuracy of 94.4%, our experimental analysis demonstrates that the suggested GNN module successfully learns graph topological information. The model's capacity to reliably identify critical, conflict-prone vertices and optimise the vertex ordering procedure for future colouring is demonstrated by its 0% false-positive rate. Our method greatly lowers the possibility of resource conflicts in 5G/6G networks and university timetabling settings by giving priority to these high-impact nodes, providing a scalable solution.

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